

LocalAI TV — Admin Dashboard & Database System

Final Specification Document · v1.0

Status:  Plan finalized after iterative review · Ready for team review and approval **Last updated:** 2026-05-14 **Owner:** Koneti Mohan Reddy (localaitv@gmail.com) **Document purpose:** Shareable spec to align founder, AI team, and development team before build begins.

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1. Executive Summary

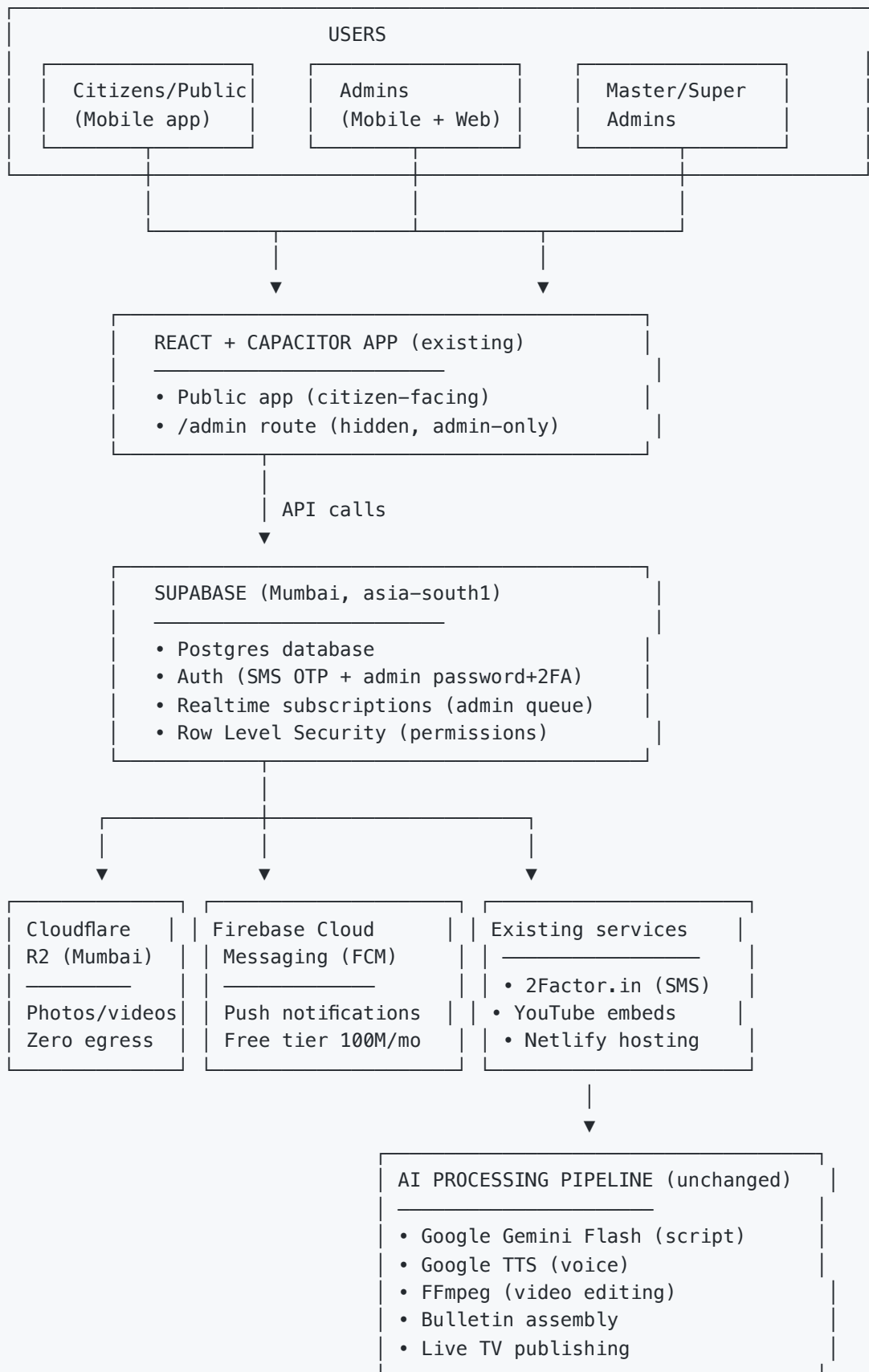
We are building an **internal moderation and content management system** for LocalAI TV that:

- Lets a 3-tier admin team (super admin → master admin → admin) review citizen submissions before they reach the live TV pipeline
- Adds a quality gate between citizen uploads and the existing AI processing pipeline (Gemini Flash → Google TTS → FFmpeg → bulletin → live TV)
- Supports state-restricted admins (a Telugu-speaking admin handles AP+TG; a Kannada admin handles Karnataka; etc.)
- Integrates Google NotebookLM-generated content (constituency news, district/state/national bulletins, debates) as a separate "auto-approved" content stream
- Provides time-slot-based publishing automation with notifications to uploaders
- Provides a comprehensive analytics dashboard with drill-down

Key non-disruption principle: The existing AI processing pipeline (script generation, TTS, video editing) stays 100% unchanged. The admin layer inserts BEFORE the AI pipeline — only approved content reaches the AI processing stage.

Monthly infrastructure cost (at launch): ₹400–2,500 **Team requirements beyond founder:** Existing AI team (unchanged) + Claude (Anthropic) for build **DevOps engineer needed:** No **Full-stack developer needed:** No (Claude builds everything)

2. System Architecture Overview



3. Three-Tier Admin System

3.1 Roles

Role	Description
Super Admin	Founder-level. Full access to everything. Only 1 person typically.
Master Admin	Senior moderator. Can create admins. Full state access. Escalates to super admin when uncertain.
Admin	Front-line moderator. Restricted to specific assigned state(s). Escalates to master admin when uncertain.
Citizen / Public	One unified role. Anyone who uploads news/content. Same registration form as before. No distinction between "citizen reporter" and "public".

3.2 Permissions Matrix

Capability	Super	Master	Admin
Create Master Admin	✓	✗	✗
Create Admin	✓	✓	✗
Suspend/remove admins below tier	✓	✓ (admins only)	✗
Geographic access:			
All states	✓	✓	✗ (restricted to assigned states)
Assigned states only	—	—	✓
Content upload (national/state/district/constituency, all 8 forms):			
Upload anywhere	✓	✓	✓ (within assigned states)
Bypass review (own uploads → directly approved)	✓	✓	✓
NotebookLM content upload:			
Constituency individual news	✓	✓	✓
District / State / National bulletins	✓	✓	✓
Debate content	✓	✓	✓
Review citizen submissions:			
Approve	✓	✓	✓
Modify & approve	✓	✓	✓
Reject with reason	✓	✓	✓
Escalate to higher tier	— (top)	→ Super	→ Master
Analytics dashboard:			
View all India stats	✓	✓	✗
View their state stats	✓	✓	✓
View reporter profiles (in their access scope)	✓	✓	✓
Audit log:			
Read all	✓	✗	✗
Read team's	✓	✓	✗

Capability	Super	Master	Admin
Read own	✓	✓	✓

3.3 State Restriction (Multi-Language Support)

When a super admin or master admin creates an Admin, they assign one or more states:

```
Create Admin form:
├ Name: Ravi Kumar
├ Phone: +91 98765 43210
├ Email: ravi@localaitv.com
└ Language access (one or more):
  ├── ☒ Andhra Pradesh (Telugu)
  ├── ☒ Telangana (Telugu)
  ├── ☐ Karnataka (Kannada)
  ├── ☐ Tamil Nadu (Tamil)
  ├── ☐ Kerala (Malayalam)
  └ ☐ Other states...
```

- **Admin** sees only content from their assigned state(s) — system auto-filters
- **Super/Master admin** sees ALL states with a dropdown to filter
- Database enforces this via Row Level Security — even if UI is bypassed, queries are filtered

4. Content Workflow (End-to-End)

STEP 1 – UPLOAD

Citizen submits via app:

- 1–3 media files (photos/videos, max 3 total combined)
- Headline (single field, voice or text, te or en)
- Description (single field, voice or text, te or en)
- Location (defaults to constituency)

- Files uploaded to Cloudflare R2 (signed URLs)
- Metadata saved to Supabase (status: pending)
- Item appears in admin queue



STEP 2 – ADMIN CLAIM (locking)

Admin clicks item from queue:

- DB locks item to that admin's ID (claimed_at = now)
- Other admins immediately stop seeing it in their queue
- Auto-expires after 30 minutes of inactivity



STEP 3 – DECISION

Admin chooses one:

- | | | |
|---|----------------|---|
| ✓ | APPROVE | – content moves to approved pool |
| ✎ | MODIFY+APPROVE | – admin edits text or removes some media, original is preserved (read-only), modified version goes to approved pool |
| ✗ | REJECT | – admin gives reason, citizen is notified, content soft-deleted (30-day retention) |
| ➡ | ESCALATE | – admin → master, master → super |

(only if approved)



STEP 4 – WEBHOOK FIRES

Supabase trigger sends webhook to your AI team's pipeline:

POST <https://your-ai-pipeline.localaitv.com/webhook/approved>

```
{  
  "content_id": "...",  
  "headline": "...",  
  "description": "...",  
  "location": "...",  
  "media_urls": [...],
```

```
"category": "news",  
"constituency": "kurnool-urban"  
}
```



STEP 5 – AI PIPELINE (your existing system, unchanged)

1. Gemini Flash generates 30–60sec script
2. Google TTS converts script to voice
3. FFmpeg edits video/photos to match voice duration
4. Original audio/music mixed where appropriate
5. Output stored in Kurnool TV public folder (or relevant)



STEP 6 – PIPELINE CALLBACK

AI pipeline calls back to our API:

POST /api/admin/mark-processed

```
{ content_id, video_url, status: 'ready_for_bulletin' }
```

- Supabase updates: status = 'ready_for_bulletin'
- Time-slot scheduler picks it up



STEP 7 – TIME-SLOT SCHEDULING

- Scheduler assigns content to next available broadcast slot
- Birthday wishes auto-scheduled for the EXACT birthday date
- Other content distributed across daily slots (8:50, 11:50, 16:50, 20:50, etc.)

10 minutes before slot:

- Uploader gets push: "Your post airs in 10 min – watch live!"
- Birthday person + wisher both notified



STEP 8 – LIVE TV BROADCAST

- Bulletin assembled (your existing 10–15 min compilation)
- Live TV broadcast on Kurnool TV channel
- Status: 'published'
- Audit log records broadcast time

5. The 8 Upload Forms — Specifications

NOTE: Field lists below are based on auditing your actual canonical app source (`286.App_v3_20260512-201203_auto-backup.jsx`). A final field-by-field audit will be done in Phase 1 build to lock every label, validation, and placeholder.

Common rules across ALL forms:

- **Media:** Maximum 3 files total (photos and/or videos in any combination)
- **Single-language fields:** Headline / description fields are ONE field each — user types/speaks in Telugu OR English; app auto-detects
- **Voice input:** Supported for text fields (Telugu or English auto-detected)
- **Location:** Defaults to user's registered constituency
- **Required confirmation:** "I confirm content is original" checkbox before submit

A. 📰 News Upload Form

Field	Type	Required	Validation
Media	Photos / Videos	✅ Yes	1–3 files, mixed allowed
Headline	Text (single field, te or en)	✅ Yes	4–30 words
Description / Details	Text (single field, te or en)	✅ Yes	20–300 words
Location	Text	Auto-filled	Defaults to constituency
Originality confirmation	Checkbox	✅ Yes	—

B. 🎂 Birthdays

Field	Type	Required
Birthday person's full name	Text	✅ Yes
Date of birth	Date	✅ Yes
Age	Auto-calculated from DOB	—
Birthday photos	Photos (max 3)	Optional
Wisher's name	Text	✅ Yes
Wisher's relation (Son / Daughter / Friend / etc.)	Text	✅ Yes
Wisher's photos	Photos (max 3)	Optional

Auto-scheduling: Submitted entries are queued to broadcast on the birthday date automatically.

C. 🏠 Marriage Days (Anniversary)

Field	Type	Required
Couple names	Text	✔ Yes
Anniversary date	Date	✔ Yes
Years of marriage	Auto-calculated	—
Couple photo(s)	Photos (max 3)	Optional
Posted by — name	Text	✔ Yes
Posted by — relation	Text	✔ Yes
Greeting message	Text	Optional

D. 💍 Upcoming Marriage

Field	Type	Required
Bride's name	Text	✔ Yes
Groom's name	Text	✔ Yes
Marriage date	Date	✔ Yes
Venue / function hall	Text	✔ Yes
Muhurtham time / Reception time	Time	Optional
Invitation card image	Photo	Optional
Invitation video	Video	Optional
Family contact phone	Phone	✔ Yes

E. 🇮🇳 Events

Field	Type	Required
Event name	Text	✓ Yes
Event type (Religious/Cultural/Political/Sports/Educational/Other)	Dropdown	✓ Yes
Date(s)	Date range	✓ Yes
Time	Time	✓ Yes
Venue / location	Text	✓ Yes
Description	Text	✓ Yes
Organizer name	Text	✓ Yes
Contact phone	Phone	✓ Yes
Photos / poster	Photos (max 3)	Optional

F. 🛍 Shopping Ad

Field	Type	Required
Item name	Text	✓ Yes
Category (Electronics/Clothing/Grocery/Furniture/Other)	Dropdown	✓ Yes
Price (₹) or "Negotiable"	Text	✓ Yes
Description	Text	✓ Yes
Photos	Photos (max 3)	✓ Yes
Seller contact phone	Phone (Call/WhatsApp)	✓ Yes
Location	Text	✓ Yes

G. 🚗 Car / Vehicle Sales

Field	Type	Required
Vehicle type (Car/Bike/Scooter/Auto/Truck/Tractor)	Dropdown	✓ Yes
Brand	Text	✓ Yes
Model	Text	✓ Yes
Year of manufacture	Year	✓ Yes
KM driven	Number	✓ Yes
Fuel type (Petrol/Diesel/CNG/Electric)	Dropdown	✓ Yes
Price (₹)	Number	✓ Yes
Owner type (1st/2nd/3rd)	Dropdown	✓ Yes
Description	Text	Optional
Seller contact phone	Phone	✓ Yes
Photos	Photos (max 3)	✓ Yes
Location	Text	✓ Yes

H. 🏠 Rentals

Field	Type	Required
Property type (House/Shop/Room/PG/Commercial/Godown)	Dropdown	✓ Yes
Configuration (1BHK/2BHK/3BHK/etc)	Dropdown	✓ Yes (if house)
Furnished (Fully/Semi/Unfurnished)	Dropdown	✓ Yes
Monthly rent (₹)	Number	✓ Yes
Security deposit (₹)	Number	Optional
Area / locality	Text	✓ Yes
Description	Text	Optional
Owner contact phone	Phone	✓ Yes
Photos	Photos (max 3)	✓ Yes

I. Jobs

Field	Type	Required
Job title	Text	 Yes
Company name	Text	 Yes
Job type (Full-time/Part-time/Contract/Daily wage)	Dropdown	 Yes
Salary range or "As per skills"	Text	 Yes
Location	Text	 Yes
Skills required	Text	Optional
Description	Text	Optional
Contact phone	Phone (Call/WhatsApp)	 Yes
Email	Email	Optional
Last date to apply	Date	Optional

Future-proofing

When you add a new category (e.g. "Who's Who"):

1. We add `'whos_who'` to the category enum
2. Define new form fields
3. Database **automatically handles them** via the JSONB `form_data` column — no schema migration needed

6. NotebookLM Integration

Google NotebookLM generates pre-processed content that bypasses both admin approval AND the AI processing pipeline. Goes directly to publishing.

Content streams from NotebookLM

Source type	Description	Duration	Geographic level
Individual news	Constituency-specific news items	30 sec – 1 min	Constituency
District bulletin	4–5 news combined into a bulletin	4–5 min	District
State bulletin	State-wide news bulletin	4–5 min	State
National bulletin	National news bulletin	4–5 min	National
Debate (any level)	Discussion-format content	Variable	Constituency / District / State / National

Upload behavior

- **Upload by:** Super Admin, Master Admin, Admin (within their assigned states)
- **Approval:** Auto-approved on upload (no review queue)
- **AI processing:** Skipped — uses raw uploaded video as-is
- **Publishing:** Goes directly into time-slot publishing queue

Upload UI

Upload NotebookLM Content

Source type:

☐ Individual constituency news

☐ District bulletin

☐ State bulletin

☐ National bulletin

☐ Debate (specify level)

Geographic level:

State: [Andhra Pradesh ▼]

District: [Kurnool ▼] (if applicable)

Constituency: [Kurnool Urban ▼] (if applicable)

Title: [...]

Description: [...]

Source video: [Choose file]

This content publishes DIRECTLY

☐ No AI processing

☐ No editing

☐ Goes to assigned time slot

[Cancel]

[Upload & Publish]

7. Database Architecture

7.1 Tech: Supabase (managed PostgreSQL) in Mumbai region

7.2 Tables

users (citizens + admins)

```
CREATE TABLE users (  
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  phone             TEXT UNIQUE NOT NULL,  
  name              TEXT,  
  email             TEXT,                                -- admin only  
  role              user_role NOT NULL,                  -- enum  
  assigned_states   TEXT[],                              -- ['AP', 'TG'] etc. NULL for super/master  
  password_hash     TEXT,                                -- admin only  
  profile_photo_url TEXT,  
  constituency      TEXT,                                -- citizen's home constituency  
  is_verified       BOOLEAN DEFAULT false,  
  created_by        UUID REFERENCES users(id),           -- who created this admin  
  suspended         BOOLEAN DEFAULT false,  
  created_at        TIMESTAMPTZ DEFAULT NOW(),  
  last_login_at     TIMESTAMPTZ  
);  
  
CREATE TYPE user_role AS ENUM (  
  'super_admin', 'master_admin', 'admin', 'citizen'  
);
```

```
CREATE TABLE content (  
  id                UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  author_id         UUID REFERENCES users(id),  
  author_role       user_role NOT NULL,      -- snapshot  
  content_source     content_source NOT NULL,  
  geography_level    geography_level NOT NULL,  
  state             TEXT,  
  district          TEXT,  
  constituency       TEXT,  
  category          content_category NOT NULL,  
  form_data         JSONB NOT NULL,          -- form fields per category  
  media_urls        JSONB,                  -- array of {type, url, thumb_url}  
  original_data     JSONB,                  -- preserved if modified  
  status            content_status NOT NULL DEFAULT 'pending',  
  claimed_by        UUID REFERENCES users(id),  
  claimed_at        TIMESTAMPTZ,  
  reviewed_by       UUID REFERENCES users(id),  
  reviewed_at       TIMESTAMPTZ,  
  rejection_reason  TEXT,  
  escalated_to      UUID REFERENCES users(id),  
  ai_processed_at   TIMESTAMPTZ,  
  ai_output_url     TEXT,  
  created_at        TIMESTAMPTZ DEFAULT NOW(),  
  updated_at        TIMESTAMPTZ DEFAULT NOW(),  
  deleted_at        TIMESTAMPTZ  
);  
  
CREATE TYPE content_source AS ENUM (  
  'citizen',          -- citizen upload (needs review + AI processing)  
  'admin_direct',     -- admin direct upload (auto-approved + AI processing)  
  'notebooklm_news',  -- NotebookLM constituency news (direct publish)  
  'notebooklm_bulletin', -- NotebookLM bulletins (direct publish)  
  'notebooklm_debate' -- NotebookLM debates (direct publish)  
);  
  
CREATE TYPE geography_level AS ENUM (  
  'national', 'state', 'district', 'constituency'  
);  
  
CREATE TYPE content_category AS ENUM (  
  'news', 'birthday', 'marriage_day', 'upcoming_marriage',  
  'event', 'shopping', 'car_sale', 'rental', 'job'  
);  
  
CREATE TYPE content_status AS ENUM (  
  'pending',          -- in admin queue  
  'in_review',        -- claimed by admin  
  'approved',         -- approved, awaiting AI processing  
  'rejected',         -- rejected by admin  
  'escalated',        -- escalated to higher tier  
  'ai_processing',    -- AI pipeline working on it  
  'ready_for_bulletin', -- AI done, ready for next time slot  
  'published',        -- aired live
```



```
'archived'          -- past content
);
```

content_audit_log

```
CREATE TABLE content_audit_log (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  content_id  UUID REFERENCES content(id),
  action      audit_action NOT NULL,
  actor_id    UUID REFERENCES users(id),
  actor_role  user_role,
  before_state JSONB,
  after_state JSONB,
  reason      TEXT,
  ip_address  INET,
  user_agent  TEXT,
  created_at  TIMESTAMPTZ DEFAULT NOW()
);

CREATE TYPE audit_action AS ENUM (
  'created', 'claimed', 'modified', 'approved', 'rejected',
  'escalated', 'unclaimed', 'restored', 'published'
);
```

publishing_queue

```
CREATE TABLE publishing_queue (
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  content_id  UUID REFERENCES content(id),
  channel     TEXT,                      -- 'kurnool-tv', 'guntur-tv', etc.
  scheduled_for_date DATE NOT NULL,
  scheduled_for_slot TIME NOT NULL,      -- '08:50:00', '11:50:00', etc.
  status      queue_status DEFAULT 'queued',
  aired_at    TIMESTAMPTZ,
  uploader_notified_at TIMESTAMPTZ,
  created_at  TIMESTAMPTZ DEFAULT NOW()
);

CREATE TYPE queue_status AS ENUM ('queued', 'aired', 'cancelled', 'failed');
```

notifications

```
CREATE TABLE notifications (  
  id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  user_id     UUID REFERENCES users(id),  
  type        notification_type NOT NULL,  
  title       TEXT NOT NULL,  
  body        TEXT,  
  payload     JSONB,  
  read        BOOLEAN DEFAULT false,  
  delivered   BOOLEAN DEFAULT false,  
  delivered_at TIMESTAMPTZ,  
  created_at  TIMESTAMPTZ DEFAULT NOW()  
);  
  
CREATE TYPE notification_type AS ENUM (  
  'content_approved', 'content_rejected', 'content_escalated',  
  'broadcast_imminent', 'broadcast_aired',  
  'new_for_review', 'admin_created', 'admin_suspended'  
);
```

constituencies (reference)

```
CREATE TABLE constituencies (  
  id          SERIAL PRIMARY KEY,  
  name        TEXT NOT NULL,  
  state       TEXT NOT NULL,  
  district    TEXT NOT NULL,  
  is_active   BOOLEAN DEFAULT true  
);  
-- Pre-populated with 175 AP + 119 TG constituencies (+ Karnataka, TN later)
```

broadcast_slots (configurable)

```
CREATE TABLE broadcast_slots (  
  id          SERIAL PRIMARY KEY,  
  slot_name   TEXT,                -- 'Morning', 'Lunch', 'Evening', 'Prime', 'Late'  
  start_time  TIME NOT NULL,       -- '08:50:00'  
  end_time    TIME NOT NULL,       -- '09:00:00'  
  is_active   BOOLEAN DEFAULT true  
);
```

7.3 Indexes

```
-- Fast admin queue queries
CREATE INDEX idx_content_queue ON content(status, state, created_at)
  WHERE status IN ('pending', 'in_review') AND deleted_at IS NULL;

-- Geography lookups
CREATE INDEX idx_content_geo ON content(state, district, constituency);

-- Approved pool by category
CREATE INDEX idx_approved_pool ON content(constituency, category, created_at DESC)
  WHERE status = 'approved';

-- Audit log lookups
CREATE INDEX idx_audit_content ON content_audit_log(content_id, created_at DESC);

-- Reporter analytics
CREATE INDEX idx_reporter_stats ON content(author_id, status, created_at);

-- Publishing queue
CREATE INDEX idx_publishing_pending ON publishing_queue(scheduled_for_date, scheduled_for_s
  WHERE status = 'queued';
```

8. Storage Folder Structure

Cloudflare R2 bucket: **localaitv-storage**

```
localaitv-storage/
├── citizen-uploads/                                ← raw files from citizens
│   ├── pending/                                    ← awaiting admin review
│   └── approved/                                  ← admin-approved, queued for AI
├── admin-direct-uploads/                          ← admin's own content (auto-approved)
├── notebooklm/                                    ← NotebookLM-generated content
│   ├── individual-news/
│   │   └── <state>/<district>/<constituency>/
│   ├── bulletins/
│   │   ├── district/<state>/<district>/
│   │   ├── state/<state>/
│   │   └── national/
│   └── debates/
│       ├── constituency/<state>/<district>/<constituency>/
│       ├── district/<state>/<district>/
│       ├── state/<state>/
│       └── national/
├── ai-processed/                                  ← output from your AI pipeline
│   └── <state>/<district>/<constituency>/<category>/
└── thumbnails/                                    ← auto-generated previews
```

Key principle: The database (Supabase) is the source of truth. Folders are just where the media files live. Database queries handle "show me approved Kurnool news" — folders provide direct file access via signed URLs.

9. Hidden Admin Access

Web access

- **Primary:** <https://admin.localaitv.com> (subdomain — not linked from public site)
- **Alternative:** <https://localaitv.com/staff> (less discoverable but works)

Mobile access

- Open the LocalAI TV app
- Profile screen → 5-tap secret sequence reveals "Staff Sign In" button (optional)
- OR: just bookmark admin.localaitv.com in mobile browser

Login flow

1. Admin opens admin.localaitv.com
2. Enters phone OR email + password
3. Server sends SMS OTP via 2Factor.in
4. Admin enters 6-digit OTP
5. Session token issued (24-hour expiry)
6. Lands on admin dashboard

For first-time admin (after being created by super/master admin):

- Receives SMS: "Welcome to LocalAI TV staff team. Login at admin.localaitv.com with phone +91XXX. Temp password: XXXX. Change on first login."
- First login forces password change + 2FA setup

10. API Endpoints

Auth (admin tier)

- [POST /api/admin/login](#) — phone+password → SMS OTP challenge
- [POST /api/admin/verify-otp](#) — confirm OTP → session token
- [POST /api/admin/logout](#)
- [GET /api/admin/me](#) — current user info + permissions
- [POST /api/admin/change-password](#)

Auth (citizen)

- [POST /api/citizen/login](#) — phone → SMS OTP (existing, unchanged)
- [POST /api/citizen/verify-otp](#) — confirm OTP

Content queue (admin-only)

- `GET /api/admin/queue?state=&category=&geo_level=` — pending items in admin's scope
- `GET /api/admin/content/:id` — full item details
- `POST /api/admin/claim/:id` — claim for review (locks)
- `POST /api/admin/release/:id` — release without deciding
- `POST /api/admin/approve/:id` — approve
- `POST /api/admin/modify-approve/:id` — modify fields, then approve
- `POST /api/admin/reject/:id` — reject with reason
- `POST /api/admin/escalate/:id` — escalate to next tier

Admin uploads

- `POST /api/admin/upload` — admin's own content (auto-approved)
- `POST /api/admin/upload-notebooklm` — NotebookLM content (auto-approved, direct publish)

Citizen uploads

- `POST /api/citizen/upload` — citizen submission (→ pending review)
- `GET /api/citizen/my-uploads` — citizen sees status of own uploads
- `GET /api/citizen/notifications` — citizen notifications

User / admin management

- `GET /api/admin/users?role=` — list admins (filtered by permission)
- `POST /api/admin/users` — create new admin (with state assignment)
- `PATCH /api/admin/users/:id` — update role, states, suspend
- `DELETE /api/admin/users/:id` — soft-delete admin

Analytics

- `GET /api/admin/analytics/dashboard?from=&to=&state=&district=&category=` — top-line stats
- `GET /api/admin/analytics/breakdown?dimension=state|district|category|reporter` — breakdown
- `GET /api/admin/analytics/reporter/:id` — individual reporter profile + lifetime stats
- `GET /api/admin/analytics/items?status=&from=&to=&...` — drill-down list

Publishing

- `GET /api/admin/publishing/queue` — view publishing schedule
- `POST /api/admin/publishing/schedule` — manually schedule item to a slot
- `POST /api/admin/publishing/cancel/:id` — cancel a scheduled broadcast

Webhooks (for AI team's pipeline)

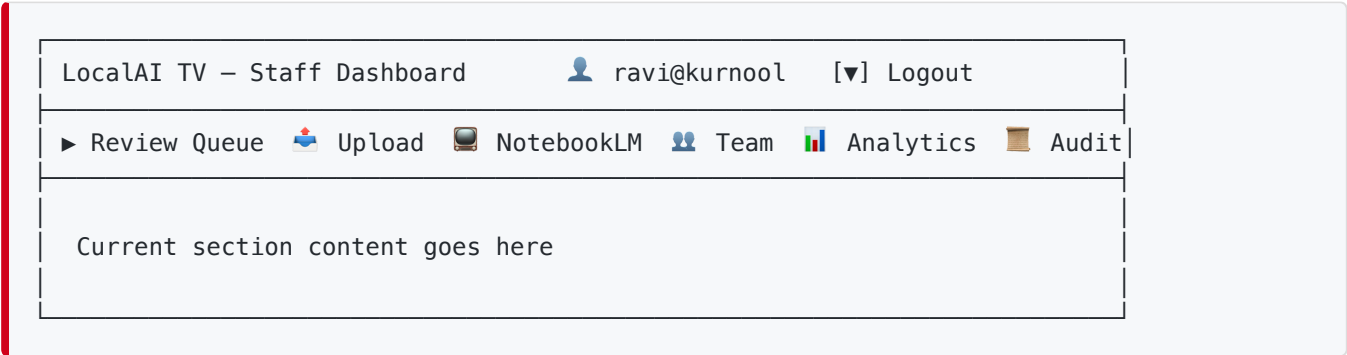
- `POST /api/webhook/content-approved` — (from us → AI pipeline) on approval
- `POST /api/webhook/ai-processed` — (from AI pipeline → us) when processing done

Audit

- GET /api/admin/audit?content_id= — audit trail for one item
- GET /api/admin/audit/recent — recent admin actions (super admin)

11. Admin Dashboard UI

Layout



Sections by tab

Tab	Visible to	Content
Review Queue	All admins	Pending citizen submissions, filterable
Upload	All admins	Admin direct upload form (auto-approved)
NotebookLM	All admins	NotebookLM content upload (direct publish)
Team	Super + Master	Manage admins, create new admins
Analytics	All admins	Dashboard, drill-down, reporter profiles
Audit	Super (all) / Master (team) / Admin (own)	Action history

(Detailed mockups in section 9 of the previous plan document.)

12. Analytics Dashboard with Drill-Down

Top-line view (super admin)

Today's Activity

[Day ▼] [Today ▼]

India Total Uploads Today: 1,247

Approved: 892 (72%) | Pending: 215 (17%) | Rejected: 94 (7%) | Processing: 46 (4%)

— BREAKDOWN (click any number to drill in) —

By State: AP 753 | TG 494

By District: Kurnool 142 | Guntur 128 | Warangal 98 ...

By Category: News 612 | Birthdays 318 | Events 156 | Cars 89 ...

By Source: Citizen 1,089 | Admin 158 | NotebookLM 0

— TOP REPORTERS TODAY —

1. Ravi Kumar (Kurnool)42 uploads · 38 approved (90%)

2. Sandhya V (Guntur)38 uploads · 35 approved (92%)

3. Praveen K (Warangal)29 uploads · 24 approved (83%)

Filters available

- **Time:** Today | Yesterday | This week | Last 7 days | This month | Last 30 days | This quarter | This year | Custom range
- **Geography:** State | District | Constituency
- **Category:** News | Birthday | Marriage | Event | Shopping | Cars | Rentals | Jobs
- **Source:** Citizen | Admin Direct | NotebookLM
- **Reporter:** Search by name

Drill-down flow

1. **Click "Approved: 892"** → list of all 892 approved items today
2. **Click "Kurnool: 142"** → list of all 142 Kurnool uploads today
3. **Click any item** → full review screen (read-only)
4. **Click reporter name** → reporter profile page

Reporter profile page



Ravi Kumar – Verified Citizen Reporter



Kurnool · Joined 2026-01-15

Lifetime stats:

Total uploads: 1,247

Approved: 1,089 (87%)

Rejected: 158 (13%)

Last upload: 12 min ago

Trend chart (last 90 days): uploads/day with approval rate overlay

By category:

News: 823 (90% approved)

Events: 214 (85% approved)

Birthdays: 98 (95% approved)

Cars: 67 (70% approved)

Recent uploads (clickable):

[list of last 20 items with status badges]

Permission scope by role

Role	View scope
Super Admin	All India
Master Admin	All India (with state filter dropdown)
Admin	Their assigned state(s) only (auto-filtered)

Future: Auto-approve based on reporter rating (Phase 2)

- When reporter sustains ≥90% approval rate over 100+ uploads → flag as "trusted"
- Trusted reporters' content auto-approved (admin can still spot-check)
- Database tracks all stats needed for this; logic added in Phase 2

13. Channel Publishing Automation & Time Slots

Broadcast slots (configurable in admin settings)

Default daily slots:

Slot Name	Time	Bulletin Length
Morning	08:50 – 09:00	~10 minutes
Lunch	11:50 – 12:00	~10 minutes
Evening	16:50 – 17:00	~10 minutes
Prime time	20:50 – 21:00	~10 minutes
Late night	22:50 – 23:00	~10 minutes

Scheduling rules

1. **General news/content:** Distributed across daily slots (round-robin, freshest first)
2. **Birthday wishes:** Auto-scheduled for slot closest to 9:00 AM on the EXACT birthday date
3. **Marriage anniversaries:** Same as birthdays — scheduled for the anniversary date
4. **Upcoming marriage announcements:** Scheduled 7 days before, 3 days before, and on the day
5. **Events:** Scheduled 2 days before, 1 day before, and on the event day
6. **Other categories (shopping, jobs, cars, rentals):** Standard distribution
7. **NotebookLM content:** Scheduled per its assigned level (national → national slot, district → district slot)

Pre-broadcast notification

10 minutes before each slot:

1. Scheduler picks content for the slot
2. Sends push notifications to all uploaders whose content is airing:

Title: 🎉 Your post airs in 10 minutes!"
Body: "Tune in to Kurnool TV at 9:00 AM. Share with friends!"
Action: Deep link to live channel

3. For birthdays: ALSO notify the birthday person + wishers

Admin override

- Super admin can manually drag/drop items between slots
- Cancel a scheduled broadcast (with audit log)
- Schedule item to specific date/slot

14. Push Notification Templates

To citizens

- **Content approved:** "🎉 Your post is approved! Airs today at [time]."
- **Content rejected:** "Your post needs changes. Reason: [admin reason]. Re-upload after corrections."
- **Broadcast imminent:** "📺 Your post airs in 10 minutes! Watch on Kurnool TV. Share with friends!"
- **Broadcast aired:** "Your post just aired on live TV! Check the replay."

To admins

- **New for review:** "New post in your queue: [category] · [constituency]."
- **Escalation (to master/super):** "[Admin name] escalated a post for your review."
- **Admin created (welcome):** "Welcome to LocalAI TV staff. Login at admin.localaitv.com. Temp password: XXXX."

To birthday person + wishers

- **Birthday broadcast scheduled:** "🎂 Your birthday wish airs on [date] at [time] on Kurnool TV. Share with family!"
- **Broadcast imminent:** "Your birthday wish airs in 10 minutes! Watch live."

All templates are editable in admin settings (super admin only).

15. AI Pipeline Webhook Integration

Non-disruption principle

Your existing AI processing pipeline (Gemini Flash → Google TTS → FFmpeg → bulletin assembly) stays **100% unchanged**. The only change for your AI team: receive a webhook instead of polling for new uploads.

Webhook contract

From admin system → AI pipeline (on approval):

```
POST https://your-ai-pipeline.localaitv.com/webhook/content-approved
Content-Type: application/json
X-LocalAITV-Signature: <HMAC for verification>
```

Body:

```
{
  "content_id": "abc-123-def",
  "headline": "కర్నూలు పై ఓవర్ ప్రారంభం",
  "description": "...",
  "location": "Kurnool City, Old Bus Stand",
  "category": "news",
  "geography_level": "constituency",
  "state": "AP",
  "district": "Kurnool",
  "constituency": "Kurnool Urban",
  "media_urls": [
    {"type": "photo", "url": "https://r2.localaitv.com/uploads/abc-1.jpg"},
    {"type": "photo", "url": "https://r2.localaitv.com/uploads/abc-2.jpg"},
    {"type": "video", "url": "https://r2.localaitv.com/uploads/abc-3.mp4"}
  ],
  "author": {
    "id": "user-xyz",
    "name": "Ravi Kumar",
    "phone": "+919876543210"
  },
  "approved_by": "admin-ravi-007",
  "approved_at": "2026-05-14T10:30:00Z"
}
```

From AI pipeline → admin system (on completion):

```
POST https://admin.localaitv.com/api/webhook/ai-processed
Content-Type: application/json
Authorization: Bearer <token>
```

Body:

```
{
  "content_id": "abc-123-def",
  "status": "ready_for_bulletin",
  "ai_output_url": "https://r2.localaitv.com/ai-processed/kurnool-urban/news/abc-123.mp4",
  "duration_seconds": 47,
  "processed_at": "2026-05-14T10:35:00Z"
}
```

What your AI team changes

```
# OLD code (their existing):
def poll_for_new_uploads():
    new_uploads = api.fetch("/uploads?status=new")
    for upload in new_uploads:
        process_with_ai(upload)

# NEW code (10-line change):
@app.route("/webhook/content-approved", methods=["POST"])
def on_admin_approved():
    # Verify HMAC signature
    if not verify_signature(request):
        return {"error": "invalid"}, 401
    content = request.json
    process_with_ai(content) # SAME function, no changes inside
    return {"ok": True}
```

That's the entire integration. Their pipeline keeps running as before.

16. Security & Row Level Security

Authentication

- **Citizens:** SMS OTP only (via 2Factor.in)
- **Admins:** Password + SMS OTP 2FA (every login)
- **Session token expiry:** 24 hours (admin), 30 days (citizen)
- **Password requirements:** Minimum 10 characters, 1 uppercase, 1 number

Row Level Security (database-enforced permissions)

```
-- Admins can only SELECT content in their geographic scope
CREATE POLICY admin_content_select ON content FOR SELECT
  USING (
    (auth.role() = 'super_admin') OR
    (auth.role() = 'master_admin') OR
    (auth.role() = 'admin' AND state = ANY(
      (SELECT assigned_states FROM users WHERE id = auth.uid())
    ))
  );

-- Only super admins can create master admins
CREATE POLICY users_create_master ON users FOR INSERT
  WITH CHECK (
    (NEW.role = 'master_admin' AND auth.role() = 'super_admin') OR
    (NEW.role = 'admin' AND auth.role() IN ('super_admin', 'master_admin')) OR
    (NEW.role = 'citizen')
  );

-- Citizens can only see their OWN content
CREATE POLICY citizen_own_content ON content FOR SELECT
  USING (author_id = auth.uid() AND deleted_at IS NULL);

-- Only super admins can hard-delete
CREATE POLICY content_delete_super ON content FOR DELETE
  USING (auth.role() = 'super_admin');
```

Even if someone bypasses the UI (direct API call), the database refuses operations they aren't authorized for.

File access security

- All media files in Cloudflare R2 are **private by default**
- Access only via **signed URLs** with 1-hour expiry
- App requests new URL each time media is viewed
- Even if a URL is leaked, it expires harmlessly

IT Rules 2021 compliance (India)

- Grievance officer contact in app (already in place)
- 24-hour response SLA on complaints
- Content takedown mechanism (built into review system)
- Audit log retained for legal record-keeping

17. Audit Log

Every action is logged with:

- WHO did it (user_id + role at time of action)
- WHAT they did (approved / rejected / modified / etc.)

- WHEN (timestamp)
- BEFORE state (snapshot of content before change)
- AFTER state (snapshot after change)
- REASON (free-text from admin)
- IP address + user agent

Retention: Indefinite (compliance), with archive option for items >1 year old.

Queryable by:

- Content ID (full history of one item)
- Actor (everything an admin did)
- Action type (all rejections, all escalations, etc.)
- Date range

18. Tech Stack Summary

Layer	Service	Cost @ launch	Cost @ 10K users
Database + Auth + Realtime	Supabase (Mumbai)	₹0	₹2,100/mo
Media storage	Cloudflare R2 (Mumbai + Chennai)	₹0 (10 GB free)	₹3,000/mo
Push notifications	Firebase Cloud Messaging	₹0	₹0
SMS OTP	2Factor.in (existing)	₹400/mo	₹2,500/mo
AI script generation	Google Gemini Flash (existing)	₹500-2,000/mo	₹5,000/mo
AI voice (TTS)	Google TTS (existing)	₹1,000-3,000/mo	₹8,000/mo
App hosting	Netlify (existing)	₹0	₹1,500/mo
Email (optional)	Resend	₹0 (free tier)	₹0
Domain	localaitv.com (existing)	₹85/mo	₹85/mo
TOTAL		₹2,000–5,500/mo	~₹22,000/mo

19. Cost Projections

Phase-wise cost during build

- Infrastructure: ₹2,000–5,500/month
- AI services (Gemini + TTS): scales with test volume — likely <₹5,000 during build
- No labor cost (founder + AI team unchanged + Claude does build)
- **Total during build: ~₹7,000–10,000/month**

Production scaling

Users (DAU)	Monthly cost
0–1K	₹2,000–5,000
1K–10K	₹5,000–22,000
10K–100K	₹22,000–1,00,000
100K–1M	₹1,00,000–5,00,000

All pay-as-you-go. Scales with actual usage.

20. Implementation Phases

Phase 1 — Foundation

- Supabase project setup (Mumbai)
- Database schema migration (all tables + indexes + RLS)
- Auth flow for admins (password + SMS OTP)
- One super admin account (you)
- Cloudflare R2 bucket setup
- Migration of any existing citizen profiles (if applicable)

Phase 2 — Admin Dashboard Core

- Dashboard layout + navigation
- Review queue with state-filtering
- Item detail / review screen
- Claim/lock mechanism
- Approve / Reject flows
- Webhook trigger to AI pipeline

Phase 3 — Admin Workflow Complete

- Modify & Approve (with original retention)
- Escalation flow (admin → master → super)
- Admin direct upload form (auto-approved)
- Push notification integration (FCM)
- Notifications to citizens on approve/reject

Phase 4 — Channel Publishing + NotebookLM

- Broadcast slots configuration
- Time-slot scheduler (cron job)
- Birthday auto-scheduling

- NotebookLM upload form (5 sub-types)
- Time-slot notifications to uploaders
- Publishing queue UI

Phase 5 — Analytics + Team Management

- Analytics dashboard with drill-down
- Reporter profile pages
- Date range filters
- Team management UI (create master/admin)
- Permission edit / suspend admin
- Audit log viewer

Phase 6 — Integration Testing + Onboarding

- AI team webhook integration testing
 - End-to-end test (citizen upload → admin review → AI process → live)
 - Onboard first batch of admins
 - User documentation
 - Final deployment to production
-

21. Team Requirements

Required:

- **Founder (Koneti Mohan Reddy)** — decisions, account setup, API keys, final approval
- **AI Team (existing)** — webhook handler addition (10 lines of code), pipeline stays unchanged
- **Claude (Anthropic AI)** — builds everything else

NOT required:

- **✗ DevOps engineer** (Supabase + R2 + Netlify are all managed services)
- **✗ Full-stack developer** (Claude builds the code)
- **✗ Database administrator** (Supabase auto-manages Postgres)
- **✗ Frontend developer** (Claude builds in React)

Optional (later):

- Part-time React developer for ongoing maintenance after launch (5–10 hours/week, useful but not urgent)

22. What's IN / OUT of v1

✓ IN v1

- Three-tier admin system (super / master / admin)
- State-restricted admin access
- Citizen review workflow (approve / modify+approve / reject / escalate)
- Locking mechanism (no double-review)
- Admin direct upload (auto-approved)
- NotebookLM upload (5 sub-types, auto-approved, direct-publish)
- Webhook integration with existing AI pipeline
- Time-slot publishing scheduler with notifications
- Push notifications (citizens + admins)
- Analytics dashboard with drill-down + reporter profiles
- Audit log
- Hidden admin access (subdomain)
- Mobile + web admin dashboards (same React/Capacitor app)

✗ OUT of v1 (future phases)

- Public analytics / leaderboards (out per founder decision)
- AI-assisted moderation (premature — wait for data)
- Reporter compensation / payout (out per founder — may add later)
- Reporter ranking-based auto-approval (database tracks stats, logic in Phase 2)
- Bulk approve / reject (wait for observed need)
- Comment moderation (no public comments yet)
- Multi-language admin UI chrome (English chrome only for v1; content is multilingual)

23. 📌 What I Need From You — Action Items

To start building immediately after your approval, please complete these registrations and share the credentials with me. **Total time on your side: ~30 minutes.**

A. Supabase (FREE — primary database + auth)

- **Where to sign up:** <https://supabase.com/dashboard/sign-up>
- **Account email:** Use localaitv@gmail.com (or your preferred work email)
- **Steps:**
 1. Click "New Project"
 2. Project name: `localaitv`
 3. Database password: Generate strong (save somewhere safe)
 4. Region: **Asia South (Mumbai)** — IMPORTANT for India latency

5. Plan: Free (will upgrade to Pro later if needed)

- **Share with me:**

- Project URL: `https://xxxxx.supabase.co`
- `anon` public key (Settings → API → anon public)
- `service_role` secret key (Settings → API → service_role)

B. Cloudflare R2 (Storage — needs credit card, but ₹0 at low usage)

- **Where to sign up:** <https://dash.cloudflare.com/sign-up>

- **Account email:** Same as Supabase

- **Steps:**

1. Sign up for Cloudflare account
2. Add a credit card (required to enable R2 — won't be charged at low usage)
3. Go to R2 → "Purchase R2 Plan"
4. Create bucket: `localaitv-storage` (Mumbai region if available)
5. Generate API token: R2 → Manage R2 API Tokens → Create API Token
6. Permissions: Object Read & Write

- **Share with me:**

- Account ID
- Access Key ID
- Secret Access Key
- Bucket name
- R2 endpoint URL

C. Firebase Cloud Messaging (FREE — push notifications)

- **Where to sign up:** <https://console.firebase.google.com/>

- **Account email:** Same Google account

- **Steps:**

1. Click "Add project"
2. Project name: `localaitv-fcm`
3. Enable Google Analytics: optional (skip if you want)
4. Once created, go to Project Settings → Cloud Messaging

- **Share with me:**

- Project ID
- Server key (Cloud Messaging tab → Server key)
- Web push certificate key (for Capacitor + web)
- Service account JSON file (Settings → Service accounts → Generate new private key)

D. Google Gemini Flash API (EXISTING — confirm access)

- **Verify here:** <https://aistudio.google.com/app/apikey>

- **Share with me:**

- API key (you already have this — confirm it works)
- OR confirm: "AI team will handle Gemini calls, don't need access"

E. Google TTS API (EXISTING — confirm access)

- **Verify here:** <https://console.cloud.google.com/apis/library/texttospeech.googleapis.com>
- **Share with me:**
 - API key or service account JSON
 - OR confirm: "AI team will handle TTS calls, don't need access"

F. 2Factor.in (EXISTING — already have key)

- **Confirm:** You shared this earlier
- **For production launch:** Complete DLT registration if not already done (₹5,000 one-time, 5-7 days)

G. Netlify (EXISTING — for deploying admin dashboard)

- **Confirm:** Use existing `localaitv@gmail.com` account
- **Action:** I'll deploy admin dashboard to a new Netlify site under your account using device-flow auth (already proven)

H. Domain DNS (admin subdomain)

- **You need to:** Point `admin.localaitv.com` to Netlify
- **Where:** Your domain registrar (GoDaddy / Namecheap / wherever you bought localaitv.com)
- **How:** Add a CNAME record:
 - Type: CNAME
 - Name: `admin`
 - Value: `<netlify-site>.netlify.app` (I'll give you the exact value during Phase 1)
- **OR:** Give me access to DNS settings and I'll configure it

I. AI Pipeline webhook URL (FROM YOUR AI TEAM)

- **Action:** Ask your AI team to provide a webhook endpoint where they'll receive approved content
- **Format:** `https://your-pipeline-domain.localaitv.com/webhook/content-approved`
- **Authentication:** I'll provide HMAC signing key for verification
- **Their effort:** ~10 lines of code to add this endpoint (replaces their current polling logic)

J. Optional — Resend (FREE — for admin email invites)

- **Sign up:** <https://resend.com/signup> (3K emails/month free)
- **Skip if:** You don't want email — SMS-only is fine

Summary of action items

Item	Time needed	Cost
A. Supabase account	5 min	₹0
B. Cloudflare R2 account (with card)	10 min	₹0 at low usage
C. Firebase project	10 min	₹0
D. Confirm Gemini API access	2 min	(existing)
E. Confirm Google TTS access	2 min	(existing)
F. 2Factor.in DLT registration (optional, for production)	5-7 days, ₹5,000	One-time
G. Confirm Netlify access	1 min	(existing)
H. DNS CNAME for <code>admin.localaitv.com</code>	5 min	₹0
I. Ask AI team for webhook URL	(their decision)	₹0
J. Resend (optional)	5 min	₹0

Total founder time: ~30-45 minutes **Total cost (one-time):** ₹0–5,000 (DLT is optional) **Total monthly cost during build:** ~₹2,000–5,500

Once you've completed A–I, paste the credentials in our chat (or save to a secure note and share via password manager). I'll start Phase 1 the moment I have them.

24. Approval & Sign-off

Founder approval

Item	Confirmed by founder?
Three-tier admin system (super / master / admin)	☐ Yes ☐ No
State-restricted admins	☐ Yes ☐ No
Form fields as specified (max 3 files, single fields)	☐ Yes ☐ No
Supabase + Cloudflare R2 architecture	☐ Yes ☐ No
Hidden admin URL (admin.localaitv.com)	☐ Yes ☐ No
Push notifications (FCM, free)	☐ Yes ☐ No
Channel publishing automation with time slots	☐ Yes ☐ No
AI Pipeline webhook integration	☐ Yes ☐ No
NotebookLM content folders (5 types)	☐ Yes ☐ No
Analytics dashboard with drill-down	☐ Yes ☐ No
3-4 week build timeline	☐ Yes ☐ No
₹2,000–5,500/month infrastructure cost	☐ Yes ☐ No
OUT of v1: payout, public leaderboards, AI moderation	☐ Yes ☐ No

AI team review

Item	Confirmed by AI team?
Webhook contract (receive POST instead of polling)	☐ Yes ☐ No
10-line code change in their pipeline	☐ Yes ☐ No
Callback URL to mark content as processed	☐ Yes ☐ No

Final approval to begin Phase 1:

Founder signature: _____ Date: _____

Contact

Founder: Koneti Mohan Reddy **Email:** localaitv@gmail.com **Company:** LocalAI Media Network Pvt Ltd
CIN: U63910KA2025PTC212593

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